# 12.20 Buffon's Needle

WORKSHEET

**The question**

**STEP I 1993, Question 16**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q16

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

**Ask yourself** The pin's centre is at distance $x$ from the nearest line, and the pin makes angle $y$ with the perpendicular to the lines. How far does the pin reach towards the line?

### STAGE 1 The integral

Covers: the integral

**Ask yourself** Substitute $t = \cos y$ in $\int_0^1 \cos^{-1} t \, \mathrm{d}t$, then integrate by parts.

### STAGE 2 Given the position

Covers: the probability given $X = x$

**Pause and try** The pin's centre is at distance $x$ from the nearest line and the pin makes angle $y$ with the perpendicular. How far towards the line does the pin reach? For fixed $x$, which values of $y$ give a crossing?

### STAGE 3 Averaging over the position

Covers: the probability for a general throw

**Ask yourself** $X$ is uniform on $[0, b]$. Integrate your conditional probability against its density, and substitute $x = at$ to use the first integral.

## What's really going on

#### Buffon's noodle

Here is a way to see the answer with almost no calculation. Bend the needle into any shape, a "noodle", and let $C$ be the number of times it crosses a line. By linearity of expectation, the expected number of crossings of a curve is the sum over its small pieces, so it is proportional to the length: $\mathrm{E}(C) = \kappa L$ for some constant $\kappa$. To find $\kappa$, take the noodle to be a circle of diameter equal to the line spacing $2b$: it always crosses exactly twice, and its length is $2\pi b$. So $\kappa \cdot 2\pi b = 2$, giving $\kappa = \frac{1}{\pi b}$. For a straight needle of length $2a < 2b$, which crosses at most once, $\mathrm{P}(\text{cross}) = \mathrm{E}(C) = \frac{2a}{\pi b}$. This argument is due to Joseph-Émile Barbier (1860), and it is a gem of probabilistic reasoning: the $\pi$ comes from the circle that was chosen to calibrate the constant.

#### Estimating $\pi$, and how not to

If $n$ needles are thrown and $c$ cross, then $\frac{c}{n} \approx \frac{2a}{\pi b}$ and $\pi \approx \frac{2an}{bc}$. The error shrinks only like $\frac{1}{\sqrt{n}}$, so to get $\pi$ to three decimal places needs about a million throws. In 1901 the Italian mathematician Mario Lazzarini claimed to have obtained $\pi \approx \frac{355}{113} = 3.1415929\ldots$ (correct to six decimal places) from 3408 throws. Statisticians have pointed out that the result is far too good to be true: with 3408 throws and a carefully chosen needle, $\frac{355}{113}$ arises from exactly 1808 crossings, and stopping the experiment at the right moment would produce it. It is a classic cautionary tale about selective reporting.

#### Buffon

Buffon was a naturalist, director of the royal botanical garden in Paris and author of a 36-volume natural history. He presented the needle problem (and a version with square tiles, the "franc-carreau" game) to the Académie des Sciences in 1733, and

published the solution in 1777. His aim was to show that geometry, not just arithmetic, could be used in probability. Integral geometry, the theory of random lines and their intersections with curves and surfaces, grew from it, and is now used in stereology, the science of estimating the size and shape of three-dimensional structures (such as cells in a tissue) from two-dimensional slices.

## Try this next

1. **A long needle.** If $a > b$, the needle may cross more than one line. Show that the expected number of crossings is still $\frac{2a}{\pi b}$.

2. **The noodle.** Bend a needle of length $2a$ into a right angle (two arms of length $a$). What is the expected number of crossings?

3. **Square tiles.** A coin of radius $r$ is tossed onto a floor of square tiles of side $s > 2r$. Find the probability that it lies entirely within one tile.

4. **Lazzarini's needle.** With $a = \frac{5}{6}b$ (so that $\frac{2a}{\pi b} = \frac{5}{3\pi}$), show that 3408 throws with 1808 crossings give the estimate $\pi \approx \frac{355}{113}$.

## Further reading


- **Daniel Klain & Gian-Carlo Rota, *Introduction to Geometric Probability*** (Cambridge University Press), which starts from Buffon's needle.
- **Lee Badger, "Lazzarini's lucky approximation of $\pi$"**, *Mathematics Magazine* 67 (1994). The statistics of a suspicious experiment.
- **Paul Nahin, *Digital Dice*** (Princeton University Press). Monte Carlo methods for probability problems.


The full guided solution is in Section 12.20:
https://beyondstep.keithmcnulty.org/ch12/buffon.html